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REVIEW

Sports performance: is there evidence that the body clock plays a role?

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Abstract Athletic performance shows a time-of-day effect, possible causes for which are environmental factors (which can be removed in laboratory studies), the sleep-wake cycle and the internal "body clock". The evidence currently available does not enable the roles of these last two factors to be separated. Even so, results indicate that the body clock probably does play some role in generating rhythms in sports performance, and that to deny this is unduly critical. Protocols to assess the separate roles of the body clock and time awake are then outlined. A serious impediment to experimental work is muscle fatigue, when maximal or sustained muscle exertion is required. Dealing with this problem can involve unacceptably prolonged protocols but alternatives which stress dexterity and eye-hand co-ordination exist, and these are directly relevant to many sports (shooting, for example). The review concludes with suggestions regarding the future value to sports physiology of chronobiological studies.

Keywords Circadian rhythms · Athletics · Muscle activity · Endogenous · Exogenous

Introduction

Most individuals consider that their athletic prowess is best in the late afternoon and early evening, and this is the time period when best performances and even world records are most often likely to be set in competitions. External factors may be in part responsible, the world records set in track-and field events in the evening reflecting the times at which Grand Prix events and major championships are held in front of large crowds and the media. However, recent reviews have considered the evidence that sports performance shows a diurnal rhythm that is, in part at least, due to the activities of a "body clock" (for example: Atkinson and Reilly 1996; Barattini 1997; Cappaert 1999; Manfredini et al. 1998; O'Connor et al. 2004; Reilly 1990, 1994; Reilly and Bambaeichi 2003; Reilly and Edwards 2007; Reilly et al. 1997a, b, 2000, 2007a, b; Waterhouse et al. 2004; Youngstedt and O'Connor 1999). Authors generally conclude that the current evidence for the role of the body clock is inconclusive, and take up a position somewhere on a continuum between "Case proved" in spite of the poor quality of much of the data and "Case not proved" because of the poor quality of the studies reported.

The present review summarises the types of evidence that are available and considers the reasons for the poor quality of data from a chronobiological perspective; methods are then suggested by which the situation can be improved in future investigations.

Measured rhythms of physical performance

Many aspects of exercise performance display a diurnal (daytime) rhythm with higher values in the late afternoon (around 1600–2000 hours) than in the morning soon after waking (around 0700–1000 hours). Examples include peak force of leg and back muscles (Coldwells et al. 1994; Guette et al. 2005; Nicolas et al. 2005, 2008a; Sedliak et al. 2007; Souissi et al. 2004; Wyse et al. 1994) and of arm muscles (Gauthier et al. 1996; Nicolas et al. 2008b), maximal

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anaerobic power output (Kin-Isler 2006; Souissi et al. 2007) and performance in broad and vertical jumps (Reilly and Down 1992). When rhythmic changes have been characterised from at least six measures obtained at equally spaced intervals throughout the 24 h, the peaks in performance are located from about 15:30 to 20:30 hours, with amplitudes ranging from 2 to 11% of the daily mean (summarised in Reilly 2007). Performance in simulated contests or time-trials also show a similarly timed diurnal rhythm, including running (Martin et al. 2001), swimming (Arnett 2002; Baxter and Reilly 1983; Martin and Thompson 2000; Reilly and Marshall 1991), cycling (Atkinson and Reilly 1995; Atkinson et al. 2005; Bessot et al. 2006; Deschenes et al. 1998; Giacomoni et al. 2006; Racinais et al. 2005a, c), and skilled tasks related to football (Reilly et al. 2007b), tennis (Atkinson and Spiers 1998) and badminton (Edwards et al. 2005b).

These activities cover a range of skills from gross locomotor functions to fine and complex tasks. The research design has mainly emphasised relatively short-duration exercise, thereby avoiding extended endurance performances where environmental factors, energy depletion, and hyperthermic fatigue become influential factors.

Problems in explaining measured rhythms of physical performance

The observation that a variable shows a circadian (about 24 h) rhythm can be explained by several mechanisms, including:

(a) External (exogenous) changes in the environment (light and temperature, "sense of occasion" from media attention and crowd behaviour).
(b) Changes related to the sleep-wake cycle (mental fatigue due to time awake and physical fatigue due to muscle activity, both decreasing performance); these might be termed "homeostatic mechanisms", reflecting the recuperative role of sleep.
(c) Internal (endogenous) changes due to the body's internal timing system (the "body clock").

Some of these factors can be illustrated by the study of Javierre et al. (1996). These authors investigated performance in 80-m sprints at several times of the day, and observed that performance normally peaked at around 1900 hours. When both the sleep-wake cycle and times of meals were either advanced or delayed by 2 h, the time of peak performance changed in a compatible direction and by a similar amount. Whatever the value of such findings, the mechanism(s) involved, especially with respect to inferences regarding possible internal or external cause(s), are unclear; they could include changes in the environment,

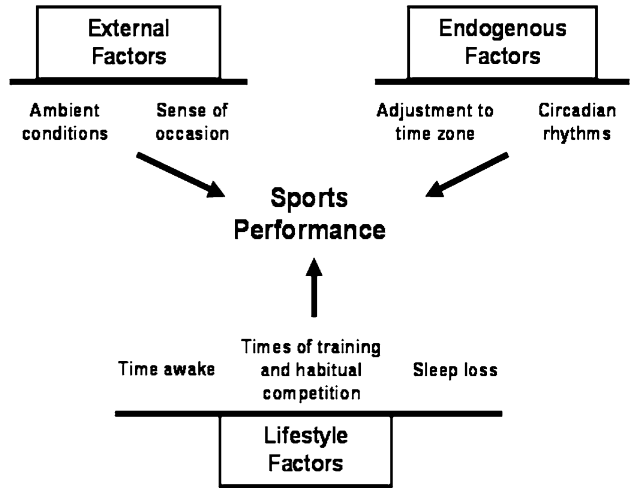

Fig. 1 Outline of factors affecting sports performance

mealtimes, the sleep–wake cycle and/or body clock. Figure 1 illustrates the effects of these mechanisms upon sports performance, but it is only a simplified description of what, in reality, can be more complex; in practice, the different factors interact with each other rather than exert their effects independently.

From the perspective of this review, one main interest is a possible role for the "body clock", a structure that consists of paired suprachiasmatic nuclei at the base of the hypothalamus. Within each cell are "clock genes" and "clock proteins" that are involved in a series of complex feedback loops (Piggins 2002). These loops produce a series of rhythmic processes that normally complete a single cycle in 24–25 h. The body clock is adjusted to an exact solar day (24 h) mainly by the light–dark cycle—light information passing from retinal ganglion cells via a direct pathway, the retinohypothalamic tract (Khalsa et al. 2003)—but the rhythmic secretion of pineal melatonin (Haimov and Arendt 1999) and, via the intergeniculate leaflet, regular changes in physical activity and "excitement" are also important.

The body clock exerts effects throughout the body via influences on temperature regulation, hormone secretion and the sleep–wake and feeding cycles; that is, the whole body behaves rhythmically with a period of 24 h.

Attempts to uncover the role of the body clock by removing "confounding factors"

To establish a possible role of the body clock, effects due to "confounding factors" (social, environmental and time-awake phenomena, see Fig. 1) need to be removed or taken into account. Methods to do this have included:

(a) Simulated field-based performance trials. These remove the "sense of occasion", but possible problems

include the difficulty of maintaining the same degree of task motivation, and the loss of "stress" that competition and associated travel engender (Iellamo et al. 2003). Effects due to the environment and the individual's sleep-wake cycle remain.

(b) Simulated laboratory-based performance trials. These remove the sense of occasion and enable environmental conditions (wind, temperature, humidity, for example) to be controlled more closely. Effects due to the individual's sleep-wake cycle remain.

(c) Measurement of components of movement and physical performance (single muscles or small groups of muscles, for example) in the laboratory, exemplifying the "reductionist" approach to a problem. For example, Bambaeichi et al. (2005) compared maximal voluntary twitches with those obtained by electrically stimulating the muscle, an approach which enabled changes due to "central" neural drive and motivation to be separated from those due to "peripheral" muscle contractile force. Again, effects due to an individual's sleep-wake cycle remain.

(d) Surrogates for performance and variables associated with exercise can be measured at rest and during exercise and recovery. Examples include responses to exercise in heart rate (Reilly and Brooks 1982), minute ventilation (Brisswalter et al. 2007; Reilly 1982), blood lactate (Forsyth and Reilly 2004) and body temperature (Aldemir et al. 2000). The results and inferences from this approach have been reviewed by Drust et al. (2005). They considered the recurrence of circadian rhythms in studies of exercise continued over 4 days and 4 nights (Reilly and Walsh 1981) as supporting the existence of an endogenous rhythm to performance. Surrogate measurements can often be performed repetitively (see below), but effects due to the sleep-wake cycle remain.

In summary, attempts to remove the effects of confounding factors have been only partially successful, and this is a reflection of the protocols that have been used. More specific protocols are required, and these will be considered in "Chronobiological methods used to separate exogenous and endogenous components".

Indirect evidence for the role of the body clock

There is a less direct way of establishing if sports performance is influenced by the body clock. One of the body clock's properties is that adjusts only slowly to a change in an individual's sleep–wake schedule. During night-work, for example, the individual attempts to work when the body clock indicates "sleep" and attempts to sleep when the body clock indicates "wakefulness", but the body clock never fully adjusts to the new sleep–wake cycle; "shift-workers' malaise" is a common phenomenon in this group.

After a time-zone transition also, the individual is initially faced with a similar problem. Consider flights across 8 time zones to the east and in the opposite direction. After the eastward flight, destination time is ahead of "body time"; the traveller does not feel tired when it is time to go to bed (local midnight) because the body clock will indicate only 1600 hours, but feels tired (midnight, body time) when the local inhabitants are waking for the next day (0800 hours, local time). After a westward flight, local time lags 8 h behind body time; the traveller begins to feel tired at about 1600 hours by the new local time and to wake up when the local inhabitants go to bed (local midnight). This mismatching between body time and local time causes symptoms of "jet lag", similar to shift-workers' malaise.

Of particular relevance to the athlete are poorer nocturnal sleep, increased daytime fatigue, and loss of motivation. The symptoms disappear as the body clock adjusts to the new time zone. While adjustment is taking place, any rhythmic effects of the body clock will be wrongly timed for the new environment; as adjustment of the body clock takes place, rhythms caused by the clock will become less synchronised to the "old" time zone and more closely synchronised to the new one. Published results for sports variables have generally not enabled this process of adjustment to be followed in detail, most of the studies suffering from one or more of the problems described in "Problems in explaining measured rhythms of physical performance" and "Attempts to uncover the role of the body clock by removing "confounding factors"".

Athletes suffer from jet lag after inter-continental flights (Reilly et al. 2005, 2007a). In some studies (for example, Reilly and Mellor 1988; Reilly et al. 2001), the rhythms of intra-aural temperature, muscle strength, acute responses to training and surrogates for physical performance were abnormally phased in the days immediately after the flight, and became normally phased to the new local time as jet-lag symptoms abated. In another study (Lemmer et al. 2002), performance at a standardised training schedule and several surrogates for physical activity were depressed after both westward and eastward inter-continental flights. Analyses of team performances and match data from American football, netball and basketball have shown that results were affected by when the teams had flown before a match (Bishop 2004; Jehue et al. 1993; Smith et al. 1997; Steenland and Deddens 1997). Decrements were most noticeable when a match was played at a time close to night-time by the visitors' home time (as would occur when a team played an evening match after travelling westwards). By contrast, an evening match could be advantageous after an eastward flight, since it would occur slightly earlier by the visitors' body time and closer to the time of peak performance.

Not all studies have shown decrements in athletic performance after time-zone transitions (Bullock et al. 2007; Hill et al. 1993; O'Connor et al. 1991). Such results might indicate the absence of an effect of an unadjusted body clock (as concluded by O'Connor et al. 2004; Youngstedt and O'Connor 1999), but there are other explanations, particularly if the frequency of measurement is too low, as is often the case in field studies. For example, consider a recording made at 1300 hours, about 4 h before the time of hypothetical peak performance. After a flight to the west across 8 time zones, a recording made at 1300 hours by local time would correspond to a time of 2100 hours (as determined by an unadjusted body clock), about 4 h after the true time of peak performance. A lack of change in performance might be wrongly interpreted to indicate that the body clock exerted no effect upon the variable being measured.

There is no substitute for recording a variable frequently in order to determine its rhythm accurately. However, frequent sampling raises a problem when all-out performance is considered—that of muscle fatigue with a corresponding inadequate recovery between time-points (see Reilly and Bambaeichi 2003). Adequate recovery does not necessarily incorporate a requirement to sleep.

Further problems when interpreting rhythms in sports performance

The above approaches indicate that physical performance in the late afternoon is better than in the early morning. Also, work-loads tend to be perceived as less arduous in the late afternoon than at other times. These changes parallel the rhythm of resting core temperature, and a causal link between temperature and physical performance has often been proposed, though not tested formally. Several groups have investigated the role of ambient temperature and warm-up upon physical performance (Arnett 2002; Atkinson et al. 2005; Racinais et al. 2004, 2005a, b), but effects of these protocols upon muscle temperature have not been established.

Even though the observed circadian rhythms in physical performance can be interpreted as evidence for a body clock promoting daytime activity, several problems need to be considered before such a role for the body clock can be accepted unreservedly. These problems include:

(a) The external validity to "sports performance" of the variables measured. Actual sports performance requires much more than activity from a single muscle or group of muscles; it also requires neural control, central decision-making and motivation.

(b) The number of times per 24 h that the variable is measured. Mathematically, the more measurements that are made, the better can a rhythm be described, and a minimum of 6 points, recorded at 4-h intervals, is minimum frequency that is acceptable. Such a protocol raises difficulties due to loss of sleep, time awake and residual muscle fatigue. If recordings are made at midnight, 0400 and 0800 hours, for example, then the midnight and 0800 hours assessments are likely to require subjects to retire late, or to rise early. For the 0400 hours test: if sleep is allowed, then "sleep inertia" (Naitoh et al. 1993) needs to be overcome before testing; if, by contrast, sleep is not allowed then there might be effects from sleep loss (below).
Whereas some authors have measured performance at 4-h intervals over the 24 h, many have compared performances at only two times of day, around 0700–1000 hours and 1700–2000 hours. The outcome of this latter protocol is a characterisation of a diurnal rather than a circadian rhythm. Chronobiologically, given that the rhythm of core temperature is a marker of the body clock, the most appropriate times are: 0300–0600 hours (temperature minimum); 0900–1200 hours (temperature rising most quickly); 1500–1800 hours (temperature maximum); 2100–2400 hours (temperature falling most quickly).

(c) "Fatigue" and the influence of the sleep-wake cycle. Fatigue has two distinct meanings in the current context: (1) the deterioration in physical performance due to preceding muscle activity; (2) the deterioration in cognitive performance due to time since waking (Waterhouse et al. 2001). Many aspects of cognitive performance are also susceptible to sleep loss (Reilly and Edwards 2007; Van Dongen and Dinges 2005), although muscle function per se seems more resilient. Effects of both types of fatigue will be considered separately in "Demonstrating that a rhythm has a clock-driven component".

Summary of problems in interpreting rhythms of physical performance

The results from the studies that have been performed indicate that a circadian rhythm is present in many measures of physical performance but they do not enable categorical statements that these rhythms can be unambiguously attributed to an internal body clock to be made. Whereas most reviewers (see, for example, Atkinson and Reilly 1996; Barattini 1997; Cappaert 1999; Manfredini et al. 1998; O'Connor et al. 2004; Reilly 1990, 1994; Reilly and Bambaeichi 2003; Reilly and Edwards 2007; Reilly et al. 1997a, b, 2000, 2007a, b; Waterhouse et al. 2004; Youngstedt and O'Connor 1999) agree that there is evidence to support such statements, all consider that more and better quality data are required. The results from some of these studies are summarised in Table 1; in this table also, the possible causes for the observed rhythms are indicated.

Table 1 Summary of investigations of circadian rhythms in sports performance, including possible causes of the rhythmicity

Site of study	Variables investigated	References	Type of measurement (possible causes of rhythmicity)
Field	Match results	Bishop (2004) Smith et al. (1997) Steenland and Deddens (1997)	Predicted changes in rhythm/time of peak performance after flights across time zones (external/sleep–wake cycle/body clock)
Laboratory	Simulated time-trials or contests	Arnett (2002) Atkinson et al. (2005) Bessot et al. (2006) Deschenes et al. (1998) Giacomoni et al. (2006) Martin and Thompson (2000) Martin et al. (2001) Racinais et al. (2005a; 2005c) Reilly and Down (1992) Reilly and Marshall (1991) Reilly and Walsh (1981)	Running, swimming, cycling and jumping (sleep–wake cycle/body clock)
	Tasks specific to particular sports	Atkinson and Spiers (1998) Edwards et al. (2005b) Reilly et al. (2007b)	Skills associated with tennis, badminton, football (sleep–wake cycle/body clock)
	Muscle groups	Coldwells et al. (1994) Gauthier et al. (1996) Guette et al. (2005) Nicolas et al. (2008a; 2008b) Sedliak et al. (2007) Wyse et al. (1994)	Leg, back and arm muscles (sleep–wake cycle/body clock)
	Single muscles	Bambaeichi et al. (2005)	Separate central and peripheral factors by direct electrical stimulation of muscle (sleep–wake cycle/body clock)
	Variables associated with performance and surrogates for performance	Aldemir et al. (2000) Brisswalter et al. (2007) Drust et al. (2005) Forsyth and Reilly (2004) Kin-Isler (2006) Reilly (1982) Reilly and Brooks (1982) Souissi et al. (2002)	Maximal aerobic/anaerobic power, maximal minute ventilation, blood lactate, core temperature, grip strength, reaction time (sleep–wake cycle/body clock)

In addition to muscle fatigue (particularly since repetitive sampling is required) and the problems already considered, there are many types of "physical performance", differing in the number of muscle groups involved, the intensity and duration of exercise, and the relative importance of muscular, neural and cognitive components. Cognitive performance is negatively affected by time-since-waking and by sleep loss (Meney et al. 1998; Van Dongen and Dinges 2005; Waterhouse et al. 2001) and so sports performance with a substantial cognitive component will be more affected by sleep loss than will performance when this component is smaller. Several studies (Bambaeichi et al. 2005; Bullock et al. 2007; Mougin et al. 1991; Souissi et al. 2003) indicate differences between types of sports performance in susceptibility to sleep loss, and loss of motivation subsequent to sleep loss has also been stressed (Blumert et al. 2007). The recent reviews by Reilly and Edwards (2007) and by Samuels (2008) can be consulted for further details of this field.

Demonstrating that a rhythm has a clock-driven component

To clarify the role of the body clock in physical performance requires protocols that, partially or completely,

separate out effects of the environment, sleep loss, time awake and muscle fatigue. If a description of the endogenous circadian rhythm is required, then this will also necessitate frequent measurement of the performance variable. Protocols that have been used to establish a role for the body clock in other physiological variables relevant to exercise (core temperature and hormone secretion, for example) will now be considered.

Chronobiological methods used to separate exogenous and endogenous components

A measured circadian rhythm has both exogenous (due to the environment and the individual's sleep-wake cycle) and endogenous (due to the body clock) components. Partial separation of these has been achieved using the following types of protocol:

(a) *Free-running protocol*: Subjects are studied in an environment from which external time cues (the environmental component) have been removed. Any rhythmicity that remains arises from combined effects of the sleep–wake cycle and the body clock, but these effects cannot be separated.

(b) *Constant routine protocol*: Subjects remain awake and sedentary for at least 24 h in a constant environment (temperature, humidity and lighting), engage in similar activities throughout (reading or listening to music), and take identical meals regularly. Effects due to the environment, sleep and waking activity are removed. Any remaining rhythmicity arises from the body clock and effects of time awake (which will be increased and so will negatively affect cognitive performance), but these two effects cannot be separated.

(c) *Ultra-short sleep–wake cycle protocol*: One version of this protocol is the "1–2 sleep–wake cycle". In this, the subjects are allowed to sleep for 1 h and then kept awake for 2 h. The environment can be kept constant and lighting adjusted in a way consonant with a 3-h "day". Identical snacks are taken each wake period. This cycle is repeated until at least 24 h have been covered. This protocol retains the stable environmental conditions required by the constant routine paradigm and reduces effects due to time awake, but some of the data are missing (when subject attempts sleep). It is also possible that sleep loss will accumulate (due to the subject's inability to fall asleep quickly at all times of the 24 h) and/or problems due to sleep inertia might become marked (if sleep is possible). This protocol has been examined comparatively little, and further evidence of its viability is required.

(d) *Forced desynchronisation protocol*: Subjects live "days" of abnormal length, equal to 21, 27 or 28 solar hours, for example. With a 28-h "day", the subjects' times of retiring, rising and eating meals, as well as all their daily activities and the imposed light–dark cycle, become 4 h later each "day". The body clock cannot adjust to such imposed schedules and so shows a circadian period just in excess of 24 h. After 7 imposed "days", the sleep–wake cycle and body clock tend to be back in synchrony (because 8×27 h = 9×24 h); this length of time is called a beat cycle. Rhythms show two components, one due to the imposed "day" (exogenous) and a circadian component due to the body clock (endogenous). Data are collected regularly throughout a beat cycle. If the results are expressed in terms of the imposed day length, the circadian component is cancelled out and the effects of time awake can be shown; if the results are expressed in terms of the endogenous period, the exogenous component is cancelled out and the circadian component becomes manifest. This method can remove effects due to the environment and separate out those due to the body clock and time awake, but it is time-consuming and arduous to perform.

(e) *Protocols involving changed sleep times*: These protocols have been used mainly with measures of cognitive performance (Åkerstedt et al. 1993, 1998; Minors et al. 1986; Waterhouse et al. 2001). Their rationale is that, if external conditions are standardised, then performance is the combined result of circadian time (which follows a sinusoid parallel to core temperature) and the homeostatic time-since-waking factor (which declines exponentially). By manipulating sleep times and then measuring performance during the course of the waking period, it is possible to calculate the mathematical functions describing these two components. To obtain a systematic mixture of circadian phases and times since waking, the protocols can be rather long (lasting over a week) and the continually changing sleep times can be disorientating for the subject. Such protocols can be seen as a variant of forced desynchronisation; they are also related to the ultra-short sleep–wake cycle protocol, which investigates rhythms when sleep times are changed and time awake is restricted to a very few hours only.

Table 2 summarises the inferences that can be drawn from various protocols if a task that does not cause muscle fatigue (enabling repetitive measurements during each wake period) is used. These protocols have formed the bedrock for studies to demonstrate the endogenous component of many circadian rhythms. Repetitive sampling is required and, when the imposed day length is greater than 24 h in the forced desynchronisation protocol, there is a small increase in time awake during the "daytimes". These factors are

Table 2 Separation of environmental, clock-driven and time-since-sleep components in chronobiological studies

Protocol	Inferences that can be drawn from results
A. Free-running	Can separate "internal" (clock-driven and time-since-sleep) and "environmental" components
	Cannot separate the "internal" components into clock-driven and time-since-sleep components
B. Constant routine	Can remove the "environmental" component and direct effects of sleep and activity
	Cannot separate the "internal" components, and the effects of time-since-sleep are increased
C. Ultra-short sleep-wake cycle	Can remove the "environmental" component and effects of time-since-sleep are greatly decreased
	Cannot remove the direct effects of sleep
D. Forced desynchronisation	Can remove the "environmental" component
	Can separate effects due to the circadian component from those due to the sleep-wake cycle
E. Changed sleep times	Can remove the "environmental" component
	Can be used to calculate effects due to the clock-driven and time-since-sleep components

These protocols require a task that does not cause muscle fatigue, so that recordings can be made at regular intervals (every 2 h, for example) throughout the waking span

unimportant with regard to core temperature, since repetitive sampling is no problem and time awake has little effect upon temperature. By contrast, for cognitive performance, repetitive sampling raises problems due to practice and/or boredom, and there is a performance decrement due to the increase in time awake causing mental fatigue. Many types of athletic performance are likely to fall between the extremes of core temperature and cognitive performance (Reilly and Edwards, 2007).

Use of "chronobiological protocols" in studies of physical performance

Very few studies using the above protocols have been performed in assessments of physical and sports performance. Early studies of free-running sleep–wake cycles employed grip strength and reaction time as surrogates for performance (Kleitman 1963). The general parallelism between these rhythms and that of core temperature led to claims that performance rhythms possessed an endogenous component, core temperature being regarded as one of the most important markers of the body clock. However, problems due to fatigue and the external validity of such rhythms were not addressed.

Three studies have been performed that deserve mention in the present context, to illustrate not only their value but also the unsolved interpretive problems.

(a) Reilly and Walsh (1981) measured heart rate and amount of activity over about 4 days in a group of subjects undergoing a "marathon" football match for charity. Subjects were allowed 5-min "comfort breaks" every hour, when they could drink and eat, but sleep was prohibited. Circadian rhythms were present in heart rate and work rate (with peaks around the late afternoon and troughs towards the end of the night), superimposed upon a downwards trend in performance and mood and an upward trend in subjective fatigue. These trends were probably due to both physical (continuous physical activity) and mental (increasing time awake) fatigue. The rhythms remaining after the removal of the trends can be considered to be in large part due to an endogenous (clock-driven) component.

(b) Callard et al. (2000) required subjects to cycle on an ergometer at a constant speed over a 24-h period, and then to stay awake and rest for a further 24 h. The maximum torque developed by voluntary isometric knee extension was estimated at 4-h intervals. Circadian rhythms peaking at about 1900 hours were observed on both days; again, additional effects of physical and mental fatigue were evident.

Both of the above studies can be considered to approximate to the requirements of a "constant routine", with a constant environment, lack of sleep and constant (or similar) amounts of physical activity throughout. The disadvantage of both of them, however, is that the two types of fatigue would have been present. It is worth noting that, if a "standard" exercise protocol with a prior resting phase is performed in the laboratory and started at different times throughout the 24 h on successive days, the resting data obtained during this protocol can approximate to those that would be obtained under constant routine conditions.

(c) Kline et al. (2007) used the "ultra-short sleep-wake cycle" protocol. Experienced swimmers were required to be awake for 2 h and then were allowed to sleep for 1 h, this 3-h sleep–wake schedule being repeated over the course of 50–55 h. Maximum performance (200-m swim) was assessed six times, at 9-h intervals, in each subject. A circadian rhythm was present, with a peak around 2300 hours (6 h before the aural temperature minimum) and a trough around 0500 hours (the temperature minimum). However, there was also a clear

effect of physical fatigue as the assessments progressed (see below).

No studies using the forced desynchronisation protocol or systematic changes in sleep times have been reported when investigating rhythms related to sports performance. Research groups appear to be reluctant to face the challenges posed by the rigours and length of these research designs.

The problem of physical fatigue in chronobiological studies

Maximal effort (often to exhaustion) will necessarily produce muscle fatigue, as will sub-maximal effort for an extended period of time. Until full recovery has taken place, performance will be compromised, with obvious implications for the repetitive performance measurements required to describe a rhythm (see comment, above, relating to the study of Kline et al. 2007).

Possible solutions to the problem of physical fatigue

To overcome the difficulty of physical fatigue, some studies have incorporated one time-point per day on successive days. For example, if recordings are made at 27-h intervals, then it is possible to obtain values separated by 3-h intervals; there is merit in making the time of collection of the last point the same as that of the first, since a lack of difference between these two points will confirm that effects of fatigue have been overcome. This protocol would require a total time of 9 days (8 × 27-h) between the first and last measurements, and the result would include also effects due to time awake.

Use of the "chronobiological methods"

More generally, the protocols considered in "Chronobiological methods used to separate exogenous and endogenous components" and Table 2 can be modified to deal with this problem of physical fatigue and allow recovery between successive measurements, but doing so lengthens the research protocol, often to an unacceptable extent. For example, assuming that (1) 3-h time-points are required to describe the 24-h circadian cycle adequately and (2) a viable measurement can be made only after at least 24 h have elapsed since the previous one, the following modifications to the protocols shown in Table 2 could be made (though the limitations of interpretation of any results from the protocols would still apply):

(a) *Free-running protocol*: One measurement could be made on consecutive subjective days, and the time of measurement could be made progressively later ("just after breakfast", "mid-morning", "around lunchtime", "mid-afternoon", and so on). It would be necessary to wake the subject for the night-time measurements. It would also be necessary to check the phase of the body clock when the measurements are made, using an established marker of it, such as core temperature.

(b) *Constant routine protocol*: After each constant routine (during which a single measurement could be made), time for recovery sleep and re-adjustment to a normal lifestyle is required. The next constant routine (a total of eight is required) is unlikely to be undertaken less than 72 h after the previous one. This protocol implies that an experiment would last about 24 days in total.

(c) *Ultra-short sleep–wake cycle protocol*: There is no *a priori* reason why this protocol cannot be continued for long periods of time, assuming that the subject does not become sleep-deprived (see above). If sleep deprivation did not occur, the "1–2 sleep–wake cycle" protocol was used and one measurement was made every 27 h, then the eight measurements would be obtained after 9 (solar) days.

(d) *Forced desynchronisation protocol*: By using imposed "days" of 27 h, a single measurement could be made, say, 1 h after waking each "day"; collecting the data during one beat cycle (9 solar days) would enable a single value corrected for circadian influences and describing performance 1 h after waking to be calculated. These results would also enable the circadian rhythm to be estimated for this part of the sleep–wake cycle (1 h after waking). A further beat cycle would be required to obtain performance 3 h after waking and corrected for circadian influences (as well as the circadian rhythm 3 h after waking). Clearly, to separate out fully the circadian rhythm corrected for time awake and the effects of time-since-sleep corrected for circadian changes would require a very long protocol. However (see Table 2), this protocol is the only one which can separate out effects of the body clock and time awake. In practice, therefore, this protocol is practicable only with tasks that do not cause muscle fatigue; if surrogates of sports performance are used, the problem of external validity must also be addressed.

(e) *Protocols involving changed sleep times*: The basic concept of manipulating sleep times to obtain a systematic mixture of circadian phases and times since waking would not need to be changed, so enabling the mathematical functions describing these two components to be deduced. However, each combination of phase and time since waking would need to be interspersed with a normal sleep–wake cycle. This would extend an already long protocol even more, although

the added "normal" sleep–wake cycles might help to reduce subject disorientation.

An alternative approach

There is an alternative approach to the problem of muscle fatigue, its relevance being that successful performance in many sports is not determined wholly by muscle strength and power. Some measures of performance (accuracy at aiming at a target, for example, see Edwards et al. 2007; 2008) have included manual dexterity and hand-eye coordination, these elements being determined by using tasks that do not cause muscle fatigue. Such tasks can be used in protocols where repetitive measurements during the course of a single day are required. Again, the results are a combination of effects produced by circadian and time-awake factors.

Continuing chronobiological studies of sport

The necessity for rigorous research designs for chronobiological studies in sport has previously been outlined by Reilly and Bambaeichi (2003), when they considered the methodological problems presented by exercise of a strenuous nature in a 24-h context and its interaction with time awake. In order to make progress, the research group must choose an appropriate task to fit the research design protocol needed for identifying circadian mechanisms. This tightening of control over prevailing environmental factors and effects of time awake does not negate the continued exploration of circadian rhythms at a descriptive or quasi-experimental level, irrespective of the relative dominance of exogenous or endogenous factors. Nevertheless, knowledge of the source of a rhythm is important when an intervention strategy is required, as in night-work or after time-zone transitions, since the endogenous component is the more resistant to an imposed phase shift.

The continued characterisation of circadian rhythms, irrespective of origin, has still a role to play in furthering knowledge of temporal aspects in sports physiology, and such studies could complement fundamental deterministic investigations. They might incorporate rhythmic aspects of different sports tasks, temporal specificity of training, and tracking physiological adjustments to circadian perturbations. Understanding the separate role played by these mechanisms would be of value to the individuals, their coaches and trainers, promoting an optimal preparation for competition and providing a rationale for improving performance. There is some evidence that training at a particular time of day produces improvements in performance that are more marked when tested at the same time of day (Edwards et al. 2005a; Hill et al. 1998; Sedliak et al. 2007; Souissi et al. 2002). The explanation of this apparent specificity of time of training is unclear (possibly an effect of the body clock and/or habituation, for example), but it might give a competitive "edge" to the individual competitor and requires further study.

There remains also the persistent problem of adequately describing effects of time awake and sleep loss. Similarly, in establishing the appropriate recovery period between consecutive time-points for different types of exercise, this period must be married to practical considerations (such as warm-ups and number of trials) to guarantee that a maximal volitional effort is obtained in each case.

Overview

Considering all the literature available, there is, on balance, sufficient evidence to indicate that many types of sports activity show circadian rhythms. To conclude that the case for rhythmicity has not been proved would seem to be overly cautious or too critical.

The cause of the observed rhythms is unclear. The fact that they can be obtained in a routine laboratory setting indicates that they are not due only to motivational and environmental factors, though these factors might be important in competition and in field conditions. The separate roles played by the body clock and the individual's sleep–wake cycle remain uncertain because the forced desynchronisation protocol and systematic manipulation of sleep times have not been used, the arduous nature of these protocols making them impracticable. In this context, therefore, the ultra-short sleep–wake cycle protocol is a promising alternative, the time-awake effect being minimal, though further validation of the protocol is required.

The timing of the circadian rhythms cannot be deduced from many of the published studies, since sampling has been too infrequent. In those cases where samples throughout the 24 h have been collected, the performance rhythms generally parallel the rhythms of core temperature (and other, potentially important rhythms—including catecholamines and other hormones, and metabolic responses). However, the hypothesis that the rhythm of temperature (or other rhythms) directly determines sports performance rhythms has not been adequately tested.

Another problem that is fundamental and more intractable is that of muscle fatigue. Effects due to muscle fatigue are an inevitable consequence of maximal or sustained effort, and this precludes repeating a test—obligatory if characteristics of a rhythm are to be established—until recovery is complete. The result is that description of a clock-driven component in a task involving maximum performance requires a very prolonged protocol. Instead, it appears simpler to investigate the presence of a clock-driven

component in a sports-related task where accuracy rather than maximal force is required.

Even if a clock-driven component is present in sports tasks, in the context of competitive performance and training, the roles of time awake, sleep loss, the environment and sense of occasion all need to be quantified. Clearly, there is still the opportunity for much more focused research in this area.

Recommendations

(a) The quasi-experimental approach (summarised in Drust et al. 2005) is one of the most practical ways of obtaining empirical data in a laboratory setting and investigating further the endogenous rhythm in performance. Nevertheless, using externally valid tasks in an experimental research design is more directly relevant.
(b) Tasks that do not produce muscle fatigue could be used to investigate several, often indirect, aspects of sports performance. The validity of these tasks needs to be established.
(c) Wider use of protocols that can separate circadian from time-awake effects—including the "ultra-short sleep–wake cycle" and forced desynchronisation protocols and systematic manipulation of sleep times—is required.

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